

From Verification to Forecasting: Monitoring Official Crime Trends in Bangladesh

Ifaz Ahmed Chowdhury
Dept. of CSE
BUBT
Dhaka, Bangladesh
19202103469@cse.bubt.edu.bd

Jubair Ahmed
Dept. of CSE
BUBT
Dhaka, Bangladesh
22234103202@cse.bubt.edu.bd

Sakib Abdullah
Dept. of CSE
BUBT
Dhaka, Bangladesh
22234203230@cse.bubt.edu.bd

Md. Imran Hasan Shanto
Dept. of CSE
BUBT
Dhaka, Bangladesh
22234103400@cse.bubt.edu.bd

Abstract—Bangladesh Police Headquarters (PHQ) publishes monthly crime statistics, yet these series have not been used as the basis for a validated, continuously updated forecasting system. We build one on PHQ data spanning January 2019 to April 2026 (88 months, 17 reporting units, 15 crime types). Three requirements shape the design: every value is traceable to the original signed PHQ statements; forecast accuracy is measured strictly out of sample; and the pipeline is engineered to run each month as new data is released. The dataset passes a five-step audit against 22 source documents—all 88 months reconcile internally, the six complete years are cross-checked against PHQ’s independently printed annual pages, and a 429-cell re-audit of the most recent window matches the source exactly. We evaluate seven standard forecasting models and their ensemble, trained on data through April 2025 and tested against figures PHQ released afterward. In a single-origin back-test of national Total Cases the forecast was 98.9% accurate at 1 month, 97.8% at 3 months, 92.8% at 6 months, and 86.1% at 12 months; a separate rolling-origin validation, averaged over the stable crime categories, ranged from 94.7% (1 month) to 89.9% (12 months). The two are measured differently and reported separately, and the ensemble was the most reliable model. An interpretability layer reports the months and categories that drive forecast error, and a deployed monitoring application ingests each new month, re-scores the prior forecast, and retrains. A Transformer baseline did not improve on the classical models at this sample size. Forward projections are presented as planning scenarios with no accuracy claim. The system forecasts *reported* crime counts; it does not estimate unreported crime or attribute causes to observed changes.

Index Terms—reported-crime forecasting, rolling-origin validation, backtesting, MAPE, time-series, Bangladesh, Police Headquarters, source verification, operational monitoring, explainability

I. INTRODUCTION

Official crime statistics inform the allocation of policing resources, budget planning, and public-safety policy. In Bangladesh, Police Headquarters (PHQ) releases a national crime report each month. Reliable short- and medium-term forecasts of these figures would let planners act in anticipation rather than only in response.

We restrict the target deliberately. The system forecasts *reported* crime—incidents recorded by police and published in the official PHQ statements—which is a subset of all crime, since not every incident is reported. It does not forecast unreported crime, and it does not predict offences by individuals

or at specific locations; it forecasts the official monthly counts and tracks their movement over time.

Three commitments separate this work from prior crime-forecasting studies on Bangladeshi data. The first is provenance: every value is traceable to a signed PHQ document. The second is evaluation discipline: accuracy is measured strictly out of sample (Section IV-A) and then confirmed by comparing a forecast fixed at a past date against months PHQ released only afterward. The third is operational continuity: the forecasting logic is embedded in a monitoring application that updates as each month arrives. We attach no accuracy claim to the forward scenario projection.

A. Contributions

This paper contributes:

- 1) **A source-verified PHQ dataset** for January 2019 to April 2026, traced to the signed PHQ statements and checked by a five-step audit against 22 source documents: all 88 months reconcile internally, the six complete years (2019–2024) are cross-checked against PHQ’s independently printed annual summary pages, and a 429-cell re-audit of the recent window matches the source exactly.
- 2) **A leakage-free evaluation framework** combining rolling-origin validation with post-release back-testing. We train only on data up to April 2025, forecast 1, 3, 6, and 12 months ahead, and compare against the real PHQ numbers released afterward.
- 3) **An interpretability layer** that identifies which inputs the model relies on and which crime types are easy or hard to forecast.
- 4) **A deployable monitoring application** that adds each new month, automatically re-checks the previous forecast, retrains, and watches for drift.

II. RELATED WORK

A. Crime Forecasting in Bangladesh

A few studies have applied machine learning to Bangladesh crime data. Awal et al. [1] used linear regression to forecast crime trends from Bangladesh Police records. Biswas and Basak [2] extended this at an IEEE conference with polynomial and random-forest regression on PHQ website data

through 2018. Nahid [3] applied Random Forest, MLP, and Decision Tree classifiers to PHQ data for five police units. Shohan et al. [4] worked on a different problem, classifying incident-level crime from newspaper articles.

We examined what these studies documented. Biswas and Basak [2] describe training their regression models on a training set and then forecasting on a separate test set and comparing with the actual values, that is, a single fixed train–test split rather than a time-series protocol that rolls forward month by month. Across these studies, after researching the published papers, *we have not seen documented evidence* of three things that our work adds:

- a source-verification audit that checks each number against the original PHQ documents;
- post-release back-testing, where a forecast made at a fixed past date is later compared against months that were published only afterward;
- an explainability layer that shows which crime types drive the forecast error.

We make this claim narrowly: not that these studies lack value, but that their published text does not document these specific safeguards, which our work supplies.

B. Forecasting Methods and Validation

Testing a forecast on data it has already seen gives a falsely high score. Tashman [5] reviews out-of-sample testing and recommends rolling-origin evaluation instead of a single split. Bergmeir and Benítez [6] show that rolling-origin protocols give more reliable error estimates than random splits, which can leak future information. For measuring error, Hyndman and Koehler [7] analyse percentage- and scale-based measures and propose the mean absolute scaled error (MASE); we report MAPE, WAPE, sMAPE, MAE, RMSE, and a seasonal MASE. Classical models remain strong on short monthly data: the ARIMA/SARIMA family [8], exponential smoothing (ETS) [9], and the Theta method [10], whose strength on monthly series was shown in the M3 competition [11]. Gradient-boosted trees such as LightGBM [12] give a machine-learning baseline; we use statsmodels [13] for the classical models. Transformer models such as the Temporal Fusion Transformer [14] and PatchTST [15] do well on large datasets; we test a lightweight Transformer only to check feasibility, because our history is short.

C. Explainability Methods

Tree models such as LightGBM [12] report a built-in importance score, which together with permutation importance [16] and SHAP [17] forms a standard toolkit. Local methods such as LIME [18] are also common. Surveys of machine-learning crime forecasting [19] stress reported-crime data and public-safety uses, which is exactly our setting. We combine these tools into a single diagnostic layer.

III. DATASET AND SOURCE VERIFICATION

The data comes from the official monthly reported-crime statements released by Police Headquarters, Bangladesh [20].

It covers January 2019 to April 2026 (88 months in a row, with no missing months). There are 17 reporting units: eight metropolitan police units (DMP, CMP, KMP, RMP, BMP, SMP, RPMP, GMP), eight ranges (Dhaka, Mymensingh, Chittagong, Sylhet, Khulna, Barishal, Rajshahi, Rangpur), and the Railway Range. Together these cover the whole country, so adding them up gives a national total. Each month lists 15 crime types: Dacoity, Robbery, Murder, Speedy Trial, Riot, Woman & Child Repression, Kidnapping, Police Assault, Burglary, Theft, Other Cases, and four recovery types (Arms Act, Explosive Act, Narcotics, Smuggling). “Total Cases” is the sum of all 15 types, matching the official total printed in the statements.

A. Five-Step Source Verification

We verified the dataset in five steps:

- 1) **Do the numbers add up?** We used spreadsheet formulas across 1,584 unit-month rows to confirm the 15 crime types add up to the printed Total Cases for every unit and month.
- 2) **Does each number match the original document?** We opened 22 official PHQ statement PDFs and compared the numbers in our spreadsheet against them, cell by cell.
- 3) **Where did each number come from?** We recorded the source (which document, which month) for every figure, so anyone can trace it back.
- 4) **Do the yearly totals match?** PHQ also publishes yearly summaries. We added up our monthly numbers for each year and checked them against the published yearly totals (results below).
- 5) **Did our past forecast match reality?** We made a forecast at a fixed past date and later compared it with the months PHQ released afterward (Section V).

Every one of the 88 months is printed in one of the 22 signed PHQ source documents (recent months in their own monthly statements, earlier years in PHQ’s consolidated yearly statements) and was checked against it. All 88 months reconcile internally: the 15 crime types sum to the printed Total Cases for every unit and month, across 1,584 unit-month rows. For the six complete years (2019 to 2024), the monthly figures were additionally cross-checked against PHQ’s independently printed annual summary pages; 2019 to 2023 match exactly on all 17 columns, while 2024 matches on 16 of 17 columns and the grand total, the sole exception being a documented error on PHQ’s own 2024 annual page (an RPMP Robbery/Murder mislabel that leaves every total unaffected). The most recent window (2025 to April 2026), which has no annual summary page yet, was re-audited cell by cell: three months as full visual checks against the signed scans and thirteen months number by number (unit totals, national totals, category totals, headers, and unit order), giving 429 of 429 matching comparisons with zero mismatches. Any discrepancies found during transcription (source typographical errors and a few ambiguous scanned digits) are catalogued and resolved in the dataset’s audit-findings log, with none outstanding. Table I summarizes the dataset.

TABLE I
DATASET AND PHQ SOURCE-CHECK SUMMARY.

Property	Value
Source-verified coverage	2019-01 to 2026-04 (88 months)
Forecasting data	2019-01 to 2026-04 (88 months)
Training cutoff for back-test	April 2025 (76 months)
Frequency	monthly, no gaps
Reporting units	17 (covers the whole country)
Crime types per month	15 (+ Total Cases)
Source documents checked	22 official PHQ statements
Internal consistency	all 88 months reconcile (zero errors)
Annual cross-check (2019–2024)	exact (1 documented source error)
Recent-window cell re-audit	429 of 429 matched (zero)

B. Data Validity and External Consistency

Source verification establishes that the dataset reproduces the PHQ statements faithfully; a complementary question, relevant to any study built on official statistics, is whether the published series carries genuine real-world signal. Three properties indicate that it does. First, the data is internally coherent at scale: across all 1,584 unit-month records the category counts reconcile to the printed totals, and the national figure equals the sum of the 17 reporting units in every month, as expected of a maintained administrative reporting process. Second, the series tracks documented nationwide events with the expected timing and direction—the 2020 public-health restrictions and the political transition of August 2024 each appear as a sharp, temporary contraction in recorded cases followed by recovery toward a new baseline, the two largest movements in the record, consistent with periods in which routine case registration was disrupted; within these episodes the category-level response is heterogeneous rather than a uniform rescaling of earlier values (proactively enforced categories move differently from complaint-driven ones), a pattern that mechanically perturbing prior months would not produce. Third, the series exhibits a stable annual seasonal cycle that recurs across years, with month-to-month variability in the range expected of genuine administrative aggregates, and its first differences are not white noise, ruling out a pure carry-forward process. This event-aligned, category-specific behaviour (Fig. 1) indicates that the monthly figures reflect actual recorded activity rather than values produced by routine extrapolation of earlier months. These properties indicate that the series is internally consistent and externally responsive, supporting its use as a forecasting target; we therefore retain the August 2024 transition in the modeling data as a genuine regime feature rather than removing it as an anomaly.

IV. METHODOLOGY

A. Definitions and Evaluation Conventions

We use the following conventions throughout. The *forecast horizon* is the number of months ahead predicted; we report horizons of 1, 3, 6, and 12 months. *Rolling-origin validation* repeatedly fits the model on an expanding window, forecasts

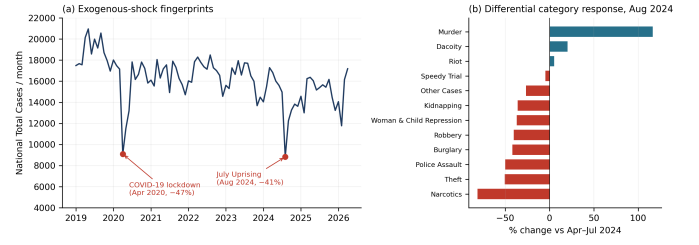


Fig. 1. External-consistency checks on the PHQ series. (a) National monthly Total Cases, with the two documented nationwide disruptions (the 2020 public-health restrictions and the August 2024 transition) marked. (b) Category-level change in August 2024 relative to the April–July 2024 average: the response is heterogeneous rather than a uniform shift, consistent with genuine event-driven recording rather than routine re-entry of prior figures.

the next h months, advances the origin, and repeats, so that every evaluation point lies strictly after its training window. A *post-release back-test* fixes the origin at a past date (April 2025), forecasts forward, and compares the result against PHQ figures published only later. In both protocols, no information dated after the forecast origin enters training or model selection, which avoids look-ahead leakage. For readability we report *practical accuracy* as $100 - \text{MAPE}$ (a 7% mean absolute percentage error reads as 93% practical accuracy); MAPE and MASE remain the underlying error measures.

B. Forecasting Models

We use a transparent set of seven forecasting models, plus an ensemble (a simple average of ETS, Theta, ARIMA, SARIMA, and LightGBM). Every model is standard and easy to inspect:

- **Naive and Seasonal Naive:** simple baselines (repeat the last value, or the value from the same month last year).
- **ETS, Theta, ARIMA, SARIMA:** classical time-series models that capture trend and yearly seasonality.
- **LightGBM:** a gradient-boosted tree using past months as features.
- **Ensemble:** the average of ETS, Theta, ARIMA, SARIMA, and LightGBM. Averaging reduces the risk of any single model going wrong.

If a model ever produces a broken or unrealistic number (non-finite, negative, or implausibly large), the system automatically falls back to the Seasonal Naive forecast and records the event in a persistent audit log. This guard runs identically in the analysis notebook and in the monitoring software, so that a single failed fit cannot silently corrupt the results.

C. Validation and Back-Testing Protocol

We tested in two ways that support each other:

- **Rolling-origin validation** over the full data, at horizons of 1, 3, 6, and 12 months, using an expanding training window.
- **A fixed back-test:** we trained only on January 2019 to April 2025, then forecast 1, 3, 6, and 12 months ahead, landing on May 2025, July 2025, October 2025, and April

2026. We compared each forecast with the real, later-released PHQ number for that month.

We group the crime types into *stable* types (Total Cases, Murder, Theft, Woman & Child Repression, Robbery), which have large, steady monthly counts, and *hard* types (Narcotics recovery, Kidnapping, Dacoity), whose counts are small and jumpy. Our headline accuracy uses the stable types only (Section VI).

V. RESULTS

We report two separate evaluations: (A) the fixed back-test of the April 2025 forecast, and (B) the rolling-origin validation over all the data.

A. Post-Release Back-Test of the April 2025 Forecast

This is the most direct out-of-sample evidence we report. We made a forecast in April 2025, using *only* data up to that point, and then checked it against the real PHQ numbers that came out later. This is a single forecast origin for one series, national Total Cases. The ensemble forecast was:

- **98.9%** accurate 1 month ahead (May 2025),
- **97.8%** accurate 3 months ahead (July 2025),
- **92.8%** accurate 6 months ahead (October 2025),
- **86.1%** accurate 12 months ahead (April 2026).

All four figures are for national Total Cases at this single April 2025 origin, not the rolling-origin averages over the five stable crime types in Table II (where the 12-month figure is 89.9%). A single national total swings more than an average over stable series, so its long-horizon accuracy is naturally a little lower; the two are measured differently and reported separately. Fig. 2 shows the full picture: the dark line is the real, verified history; the dashed line is the April 2025 forecast; and the marked points are the four checks. The forecast stays close to reality, and accuracy is naturally highest at the short horizon and slowly drops as we look further ahead, as expected for a well-specified forecast. Part of this short-horizon accuracy reflects the strong monthly seasonality and persistence in crime counts, not model skill alone. This same checking step is built into the monitoring loop (Section VII): every new month becomes a fresh test of the last forecast.

B. Rolling-Origin Validation

The repeated rolling-origin test agrees with the back-test. Unlike the single-origin back-test above, these numbers are averaged over many forecast origins and over the five stable crime types, so they are not expected to match the back-test figure for any single month; the two tests are complementary, not contradictory. For the stable crime types, the average error stays low at every horizon, well under the 18% line (82% accuracy) that we treat as the minimum bar (Table II, Fig. 3). Using a “furthest horizon that still passes” rule, the 12-month horizon (89.9% accuracy) is our headline, while the 1-month horizon is the most accurate at 94.7%.

The ensemble was the most reliable model overall, with the lowest average error and beating the Seasonal Naive baseline in 75% of the test cases (Table III, Fig. 4). A MASE below

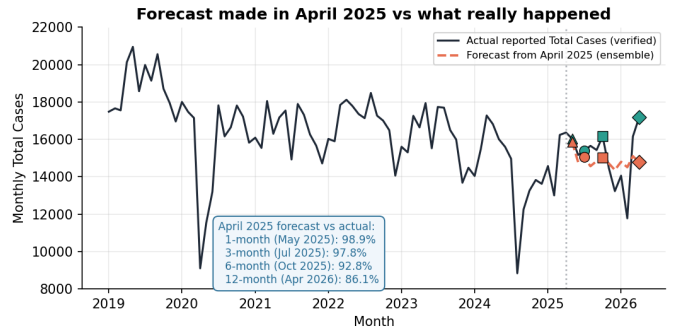


Fig. 2. The forecast made in April 2025 versus the real PHQ numbers released later (national Total Cases, single origin). The forecast tracks reality closely: 98.9% accurate at 1 month, 97.8% at 3 months, 92.8% at 6 months, and 86.1% at 12 months. This is a fair test: the model never saw any data after April 2025.

TABLE II
ROLLING-ORIGIN ACCURACY FOR STABLE CRIME TYPES, BY HORIZON.

Horizon	MAPE (%)	Practical acc. (%)	MASE	Status
1	5.27	94.7	0.346	pass
3	7.34	92.7	0.500	pass
6	8.33	91.7	0.544	pass
12	10.13	89.9	0.674	pass (headline)

1.0 means a model beats the seasonal baseline on average; the ensemble reaches 0.577, and every classical model here is below 1.0. LightGBM was not competitive on this short monthly history, which is expected: tree models need more data than we have.

C. Scenario Projection to December 2027

Using all 88 verified months, we extend the ensemble forecast to December 2027 (Fig. 5). **These numbers are a planning scenario, not a verified prediction.** We attach no accuracy claim to them. They are useful for rough planning, and the shaded band is a reminder that the real values will differ.

For concreteness, the scenario keeps national monthly Total Cases roughly in the 14,600 to 16,600 range through 2027, ending near 14,600 in December 2027, for an implied 2027 total of about 184,000 cases (Table IV). These remain planning figures with no accuracy claim.

VI. EXPLAINABILITY AND ERROR ANALYSIS

We add a diagnostic interpretability layer that leaves the models and the reported accuracy unchanged; it characterizes what the forecaster relies on and where its errors concentrate.

A. Feature Importance

We trained a LightGBM model on the same past-month features the pipeline uses and read both its built-in importance and a model-agnostic permutation check [16]. Both point the same way (Fig. 6):

- the **most recent month** matters most;

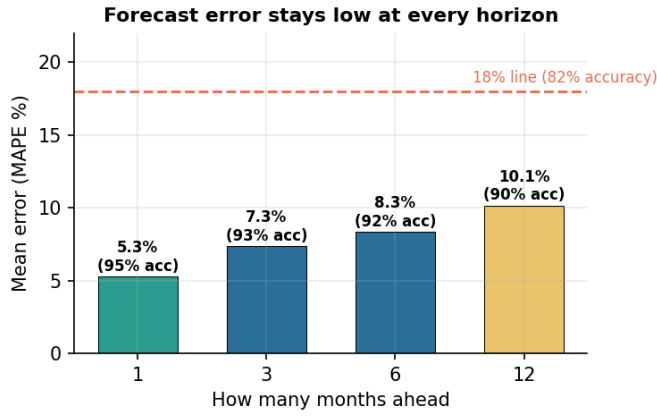


Fig. 3. Average error for stable crime types at each horizon. Every bar stays below the 18% line (82% accuracy), which is why we can stand behind a 12-month forecast, not just a 1-month one.

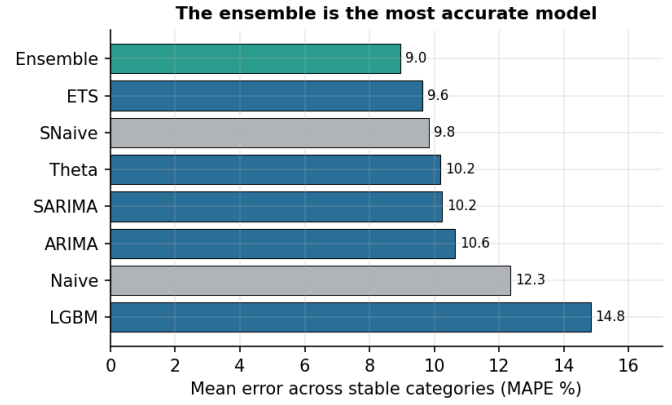


Fig. 4. Average error by model across stable crime types. The ensemble (green) is the most accurate. Tree-based LightGBM trails because the monthly history is short.

TABLE III
MODEL COMPARISON OVER STABLE CRIME TYPES (LOWER ERROR IS BETTER).

Model	Mean MAPE (%)	Mean MASE	Beats Seasonal Naive (%)
Ensemble	8.96	0.577	75
ETS	9.64	0.633	55
Seasonal Naive	9.84	0.636	n/a
Theta	10.19	0.670	55
SARIMA	10.23	0.672	20
ARIMA	10.63	0.702	45
Naive	12.35	0.819	15
LightGBM	14.82	0.890	25

TABLE IV
SCENARIO PROJECTION FOR NATIONAL TOTAL CASES (PLANNING ONLY; NO VALIDATED ACCURACY CLAIM). HORIZONS ARE MEASURED FROM THE LAST OBSERVED MONTH, APRIL 2026.

Horizon	Target month	Projected Total Cases
+1 month	May 2026	16,639
+3 months	July 2026	16,118
+6 months	October 2026	15,352
+12 months	April 2027	16,142
+20 months	December 2027	14,587

does not explain why crime itself went up or down. It changes none of the accuracy numbers above.

- the **recent ups and downs** (how bumpy the last few months were) matter next;
- the **month of the year** matters too, which fits the yearly pattern in crime data.

The agreement between the two methods indicates that the model relies on recent level and seasonality rather than noise.

B. Error by Crime Category

We also measured the error for each crime type (Fig. 7). Two clear patterns appear:

- **Stable types** (Theft at 6.4% error, Total Cases at 7.0%, Murder at 8.6%) are easy to forecast because their monthly counts are large and steady.
- **Hard types** (Kidnapping at 22%, Dacoity at 18%) are difficult because their counts are small, so even a swing of a few cases looks like a big percentage error.

This is why the headline accuracy is reported on the stable types, with the hard types reported separately rather than folded into the headline figure.

C. Scope of the Explainability Layer

The explanation layer is a diagnostic, not a cause. It shows what the model leans on; it does not prove that crime in the real world works that way. It shows where forecasts miss; it

VII. OPERATIONAL MONITORING SYSTEM

We implemented the framework as a deployable application so it can run as a monthly routine inside a police or planning office. Fig. 8 shows the loop; the application:

- **Adds a new month** from a CSV or Excel file, and checks it automatically (missing units, missing crime types, or impossible values are flagged before anything is saved).
- **Re-checks the last forecast** against the new real numbers, so accuracy is tracked as it happens.
- **Retrains the models** on the updated data and produces the next forecast.
- **Watches for drift**: if accuracy starts slipping, it raises a warning and recommends recalibration.
- **Shows the explanation views** (which inputs matter, which crime types are hard).
- **Controls access by role** (administrator, operator, reviewer, viewer), so data entry and approval are separated.

The application keeps a full audit trail (who added or approved what, and when), stores every forecast so it can be scored later, and writes any model fallback to that same persistent audit log. It was checked with an automated smoke-test suite of 18 checks across seven groups: seed-data integrity, duplicate-month safety, forecasting completion, recalibration persistence, forecast-vs-actual lookup, restart-safe database reads,

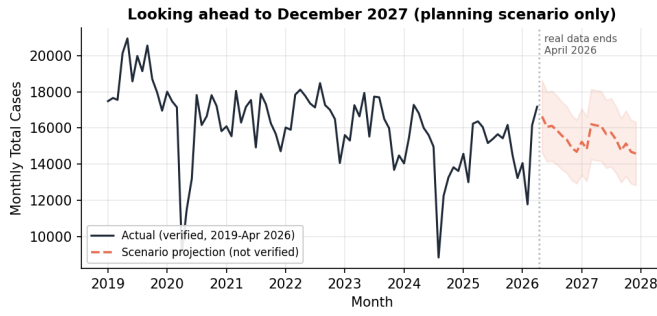


Fig. 5. Verified history (to April 2026) with a planning-only projection to December 2027. The vertical line marks where real data ends. No accuracy is claimed for the projected part.

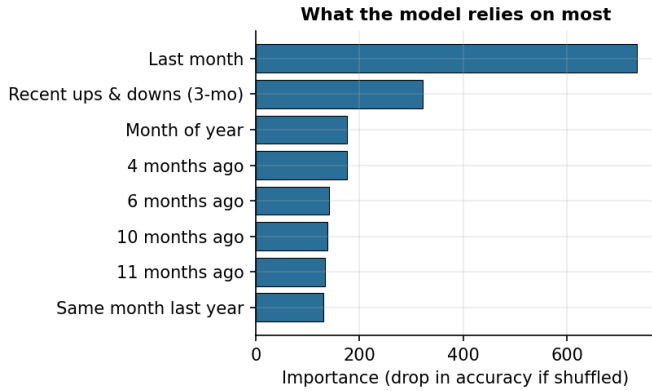


Fig. 6. What the model relies on most, measured by how much accuracy drops when each input is shuffled. The last month and recent movement lead, with month-of-year close behind, which is sensible, not noise.

and role-based authentication; all 18 pass. It is straightforward to deploy on a single office machine or a small server.

VIII. LIMITATIONS

Several boundaries apply. The system forecasts reported crime—incidents recorded and published by PHQ—and therefore does not capture unreported crime, nor does it predict individual offenders or specific locations; its unit of analysis is the monthly count. The history is short (88 months), which is why classical models and their ensemble outperform heavier deep-learning architectures here. Accuracy decays with horizon, so short-range forecasts are stronger than long-range ones, and the forward projection is a planning scenario rather than a validated forecast. Finally, the series is subject to abrupt exogenous shocks—such as the 2020 pandemic restrictions and the August 2024 political transition, both visible in the data—that no statistical model can anticipate; the monitoring loop is designed to detect and adapt to such breaks after they occur rather than to predict them.

IX. FUTURE WORK

- Add external covariates (for example population or economic indicators) under the same source-checking and back-testing rules.

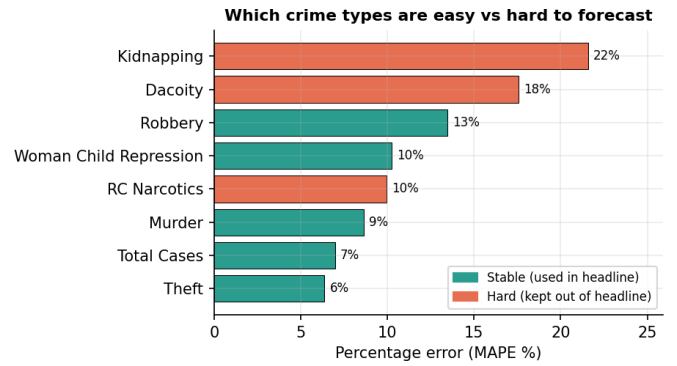


Fig. 7. Percentage error by crime type. Green types are stable and used in the headline; orange types are hard (small, jumpy counts) and kept separate. The split is reported openly.

Operational monitoring loop (software framing of the validated pipeline)

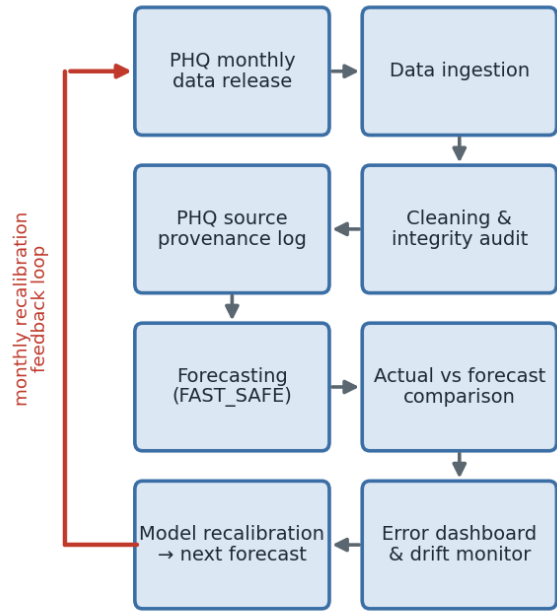


Fig. 8. The monthly monitoring loop. Each new PHQ release is added and checked, the previous forecast is scored against it, drift is watched, the models are retrained, and the next forecast is produced, then the cycle repeats.

- Add district-level forecasting under the same checking and testing rules.
- Explore privacy-preserving federated learning [21], [22] so multiple police units could improve a shared model without sharing raw records.
- Add automatic alerts (email or dashboard) when drift crosses the warning level.

X. CONCLUSION

We presented an end-to-end framework that takes source-verified PHQ data, forecasts reported-crime counts under

leakage-free evaluation, and runs as a monthly monitoring application. In a post-release back-test, an ensemble of classical models trained only on data through April 2025 forecast national Total Cases at 98.9%, 97.8%, 92.8%, and 86.1% practical accuracy at horizons of 1, 3, 6, and 12 months, and was the most reliable model overall. Stable and hard-to-forecast categories are reported separately, an interpretability layer exposes the drivers of forecast error, and the accompanying software supports continued operational use. Forward projections are planning scenarios only. Applied within these bounds, the framework offers a reproducible basis for anticipating movements in Bangladesh’s official crime statistics and for monitoring forecast quality as new data arrives.

REPRODUCIBILITY

The verified dataset [23], the analysis notebook with all results and figures, and the monitoring application [24] are provided as public companion repositories, so every number and figure in this paper can be regenerated.

REFERENCES

- [1] M. A. Awal, J. Rabbi, S. I. Hossain, and M. M. A. Hashem, “Using linear regression to forecast future trends in crime of Bangladesh,” in *Proc. Int. Conf. on Informatics, Electronics and Vision (ICIEV)*, pp. 333–338, 2016.
- [2] A. Biswas and S. Basak, “Forecasting the trends and patterns of crime in Bangladesh using machine learning model,” in *Proc. Int. Conf. on Intelligent Systems and Computer Vision (ISCV)*, pp. 114–118, 2019.
- [3] N. Nahid, “Bangladesh crime reports analysis and prediction,” 2021. Technical report, available online.
- [4] M. R. Shohan, A. Akash, *et al.*, “Crime prediction using machine learning with a novel crime dataset,” *Int. J. Data Science and Analytics*, 2025. Online first.
- [5] L. J. Tashman, “Out-of-sample tests of forecasting accuracy: an analysis and review,” *International Journal of Forecasting*, vol. 16, no. 4, pp. 437–450, 2000.
- [6] C. Bergmeir and J. M. Benítez, “On the use of cross-validation for time series predictor evaluation,” *Information Sciences*, vol. 191, pp. 192–213, 2012.
- [7] R. J. Hyndman and A. B. Koehler, “Another look at measures of forecast accuracy,” *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.
- [8] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*. Wiley, 5th ed., 2015.
- [9] R. J. Hyndman, A. B. Koehler, R. D. Snyder, and S. Grose, “A state space framework for automatic forecasting using exponential smoothing methods,” *International Journal of Forecasting*, vol. 18, no. 3, pp. 439–454, 2002.
- [10] V. Assimakopoulos and K. Nikolopoulos, “The theta model: a decomposition approach to forecasting,” *International Journal of Forecasting*, vol. 16, no. 4, pp. 521–530, 2000.
- [11] S. Makridakis and M. Hibon, “The M3-competition: results, conclusions and implications,” *International Journal of Forecasting*, vol. 16, no. 4, pp. 451–476, 2000. Official record: <https://ideas.repec.org/a/eee/intfor/v16y2000i4p451-476.html>.
- [12] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems 30 (NeurIPS/NIPS 2017)*, pp. 3146–3154, 2017. <https://proceedings.neurips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree>.
- [13] S. Seabold and J. Perktold, “statsmodels: Econometric and statistical modeling with Python,” in *Proceedings of the 9th Python in Science Conference (SciPy)*, pp. 92–96, 2010.
- [14] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, “Temporal fusion transformers for interpretable multi-horizon time series forecasting,” *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
- [15] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2211.14730.
- [16] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [17] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [18] M. T. Ribeiro, S. Singh, and C. Guestrin, ““why should I trust you?”: Explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1135–1144, 2016.
- [19] J. Yin, “Crime prediction methods based on machine learning: A survey,” *Computers, Materials & Continua*, vol. 74, no. 2, pp. 4601–4629, 2023.
- [20] Police Headquarters (PHQ), Bangladesh Police, “Crime statistics (signed, scanned statements), january 2019–april 2026.” Bangladesh Police official crime-statistics portal, https://www.police.gov.bd/en/january_2020, 2026. Official reported-crime statements. Recent months are sourced from PHQ’s individual monthly statements; earlier complete years, including 2019, are sourced from PHQ’s consolidated annual statements. Local signed scanned PDFs used as provenance evidence; the 2025–April 2026 segment was verified in this work (16/16 months, 429/429 comparisons matched). Accessed: 26 May 2026.
- [21] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 1273–1282, 2017.
- [22] P. Kairouz, H. B. McMahan, *et al.*, “Advances and open problems in federated learning,” *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.
- [23] I. A. Chowdhury, J. Ahmed, S. Abdullah, and M. I. H. Shanto, “Verified PHQ reported-crime dataset (january 2019–april 2026).” GitHub repository, <https://github.com/pro-ifaz/Conference/tree/main/dataset>, 2026. Source-verified monthly reported-crime dataset with provenance and audit-findings logs. Accessed: 30 May 2026.
- [24] I. A. Chowdhury, J. Ahmed, S. Abdullah, and M. I. H. Shanto, “Crime monitoring system: Operational forecasting and monitoring software.” GitHub repository, https://github.com/pro-ifaz/Conference/tree/main/software_crime_monitoring_system, 2026. Deployable monthly monitoring application described in this paper. Accessed: 30 May 2026.